Northern Quahog (*Mercenaria mercenaria*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	3.96	3	
Complexity in Reproductive Strategy	2.28	2.78	
Dispersal of Early Life Stages	2.52	2.64	
Egg and Larval Survival and Settlement Requirements	1.8	2.6	
Historical Variability in Temperature	1.12	2.6	
Juvenile Habitat Requirements	1.36	2.65	
Other Stressors	2.92	2.88	
Population Growth Rate	3.08	2.72	
Prey Specificity	1.36	2.3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.84	2.5	
Spawning Cycle	2.88	2.86	
Stock Size/Status	2.28	0.84	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.41	1.32	
Bottom Oxygen	2.28	1.33	
Bottom Salinity	3.23	1.11	
Bottom Aragonite Solubility	1.13	0.45	
Net primary production	1.46	0.69	
Downward Flux of Particulate Organic Carbon	1.6	0.73	
Total Exposure	**Moderate (2.31)**	**1.17**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (6.94)**	**3.51**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Northern Quahog. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Very High	High	Decrease
Exposure	High	Moderate	Decrease
Vulnerability	Very High	Moderate	Decrease
Directional Effect	Negative	Negative	None

Stocks

Northern Quahog is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

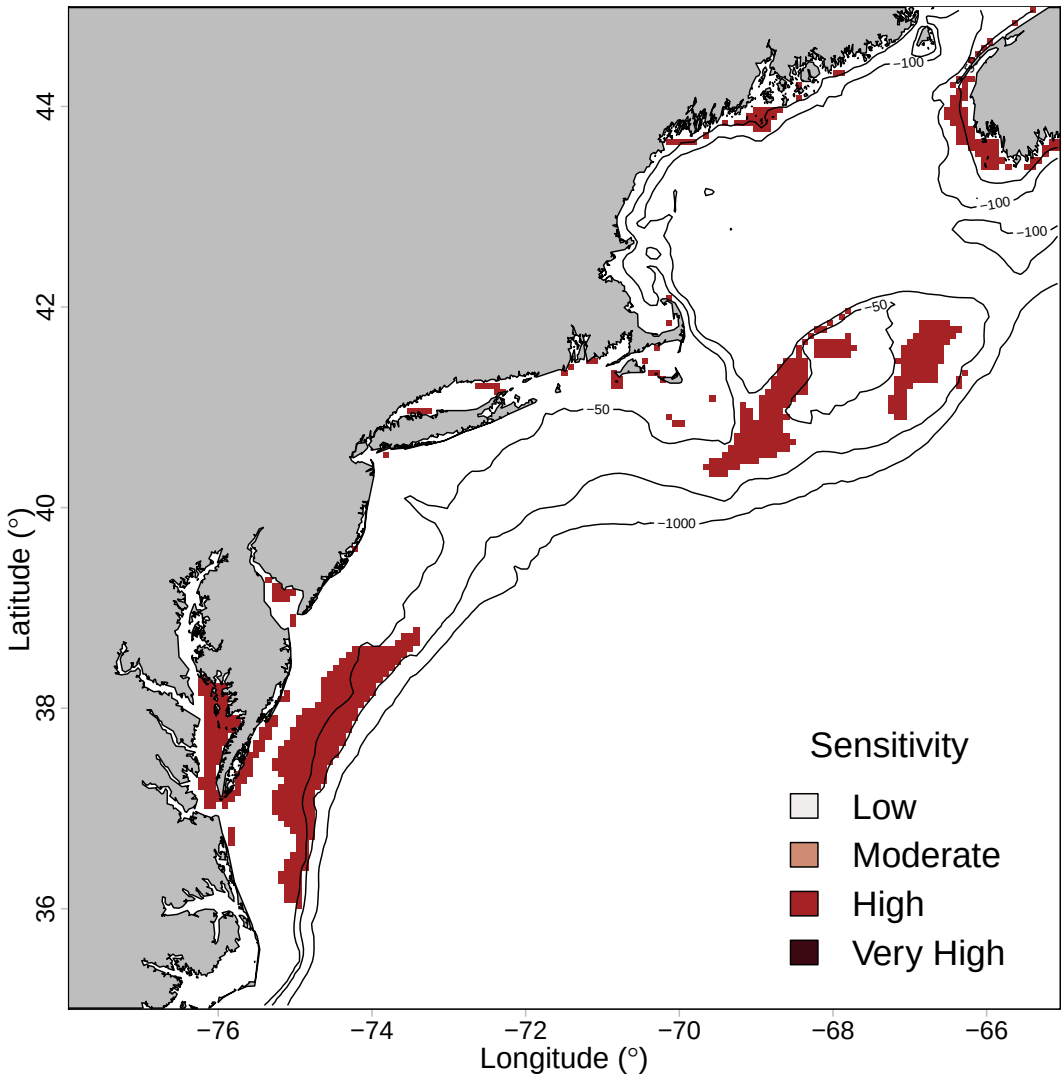


Figure 2: Northern Quahog Sensitivity on the Northeast Shelf

The Northern Quahog received a **High** Sensitivity score, with 100% certainty. Three attributes received scores greater than 3: Adult Mobility (4), Relationship to Calcified Organisms (3.8), and Population Growth Rate (3.1) (Figure 1).

Northern Quahogs, like other bivalves, are mostly sessile, only capable of small-scale movements (Stanley and DeWitt 1983). Similarly, they also have calcium carbonate shells that put them at risk of ocean acidification (Stanley and DeWitt 1983). They are long-lived, slowly maturing species, reaching maturity at 1-3 years and maximum ages usually around 12-20 years, although 46 year old quahogs have been observed (Bricelj 1992) (Ridgway et al. 2011).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Northern Quahogs received a Very High sensitivity score. This was driven by Sensitivity to Ocean Acidification, Adult Mobility, and Other Stressors (Hare et al. 2016). In our analysis, Other Stressors received a score of 2.9 (Figure 1).

Exposure:

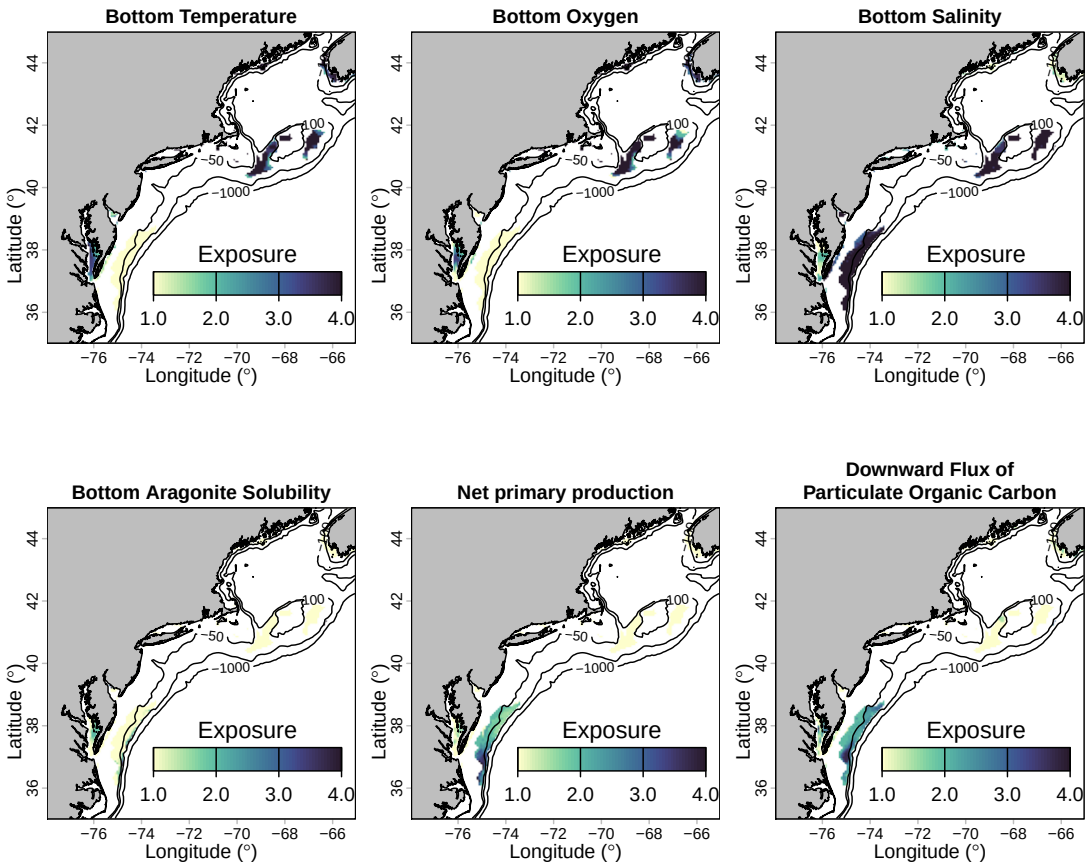


Figure 3: Maps of each Exposure Factor for Northern Quahog. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for the Northern Quahog was **Moderate** (2.31 +/- 1.17) (Figure 1). Exposure was Very High along Georges Bank and Low off the coast of the DelMarVa Peninsula (Figure 4). These patterns were driven by bottom temperature, oxygen, and salinity exposures (Figure 3).

In the previous assessment, Northern Quahog received a High Exposure score (Hare et al. 2016). This score was driven by high sea surface temperature, air temperature, and ocean acidification exposure scores (Hare et al. 2016).

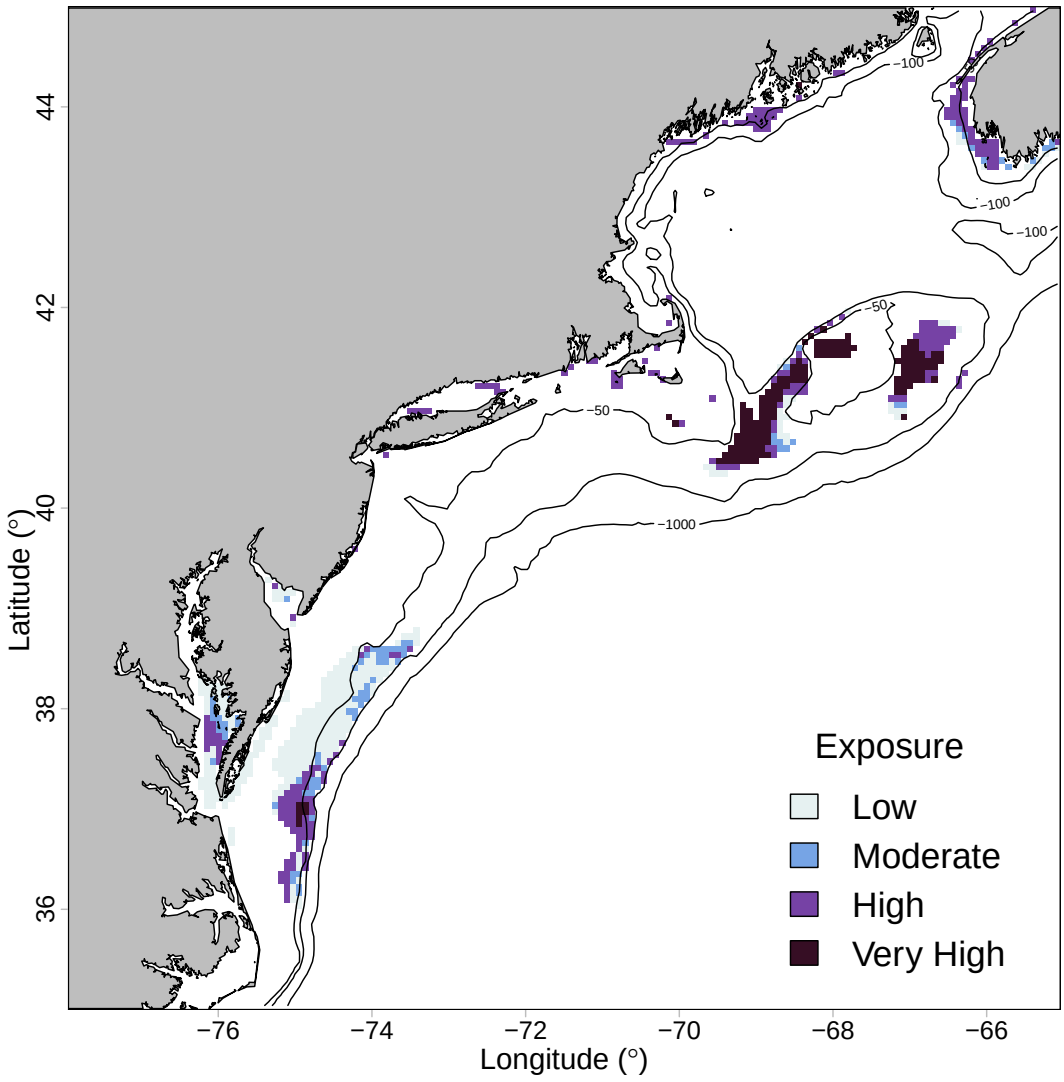


Figure 4: Total Exposure on the Northeast Shelf for Northern Quahog. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

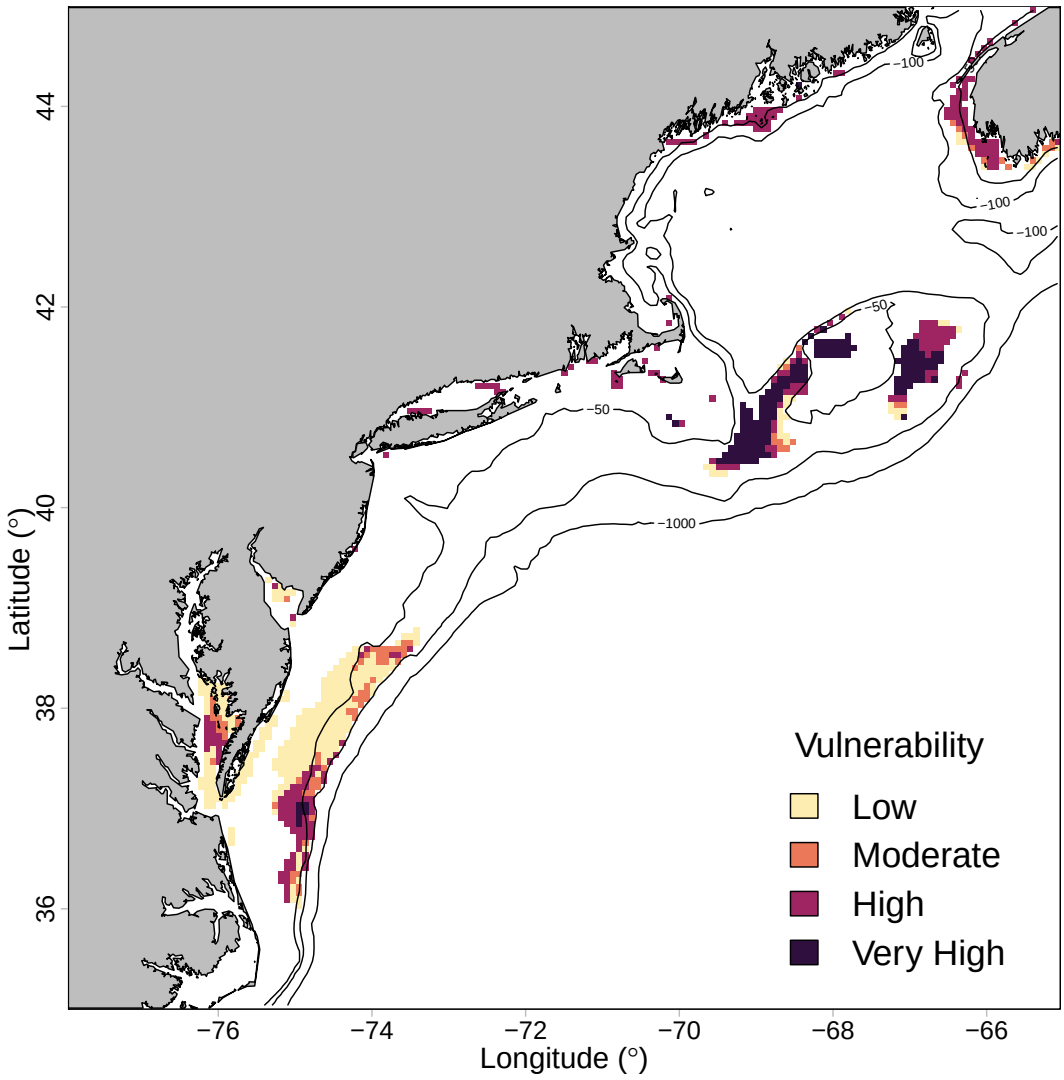


Figure 5: Vulnerability on the Northeast Shelf for Northern Quahog. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Northern Quahog was **Moderate** (6.94 +/- 3.51) (Figure 1). This was driven by Very High vulnerability scores on Georges Bank (Figure 5).

In the previous Assessment, Northern Quahog received a Very High Vulnerability score, driven by Very High sensitivity scores (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality. Only Stock Size/Status was considered a data gap (mean attribute data quality < 2) (Figure 1).

Model Confidence Score:

The Northern Quahog ensemble model received a 0.5 (+/- 0.5) mean model confidence score, indicating experts had no confidence in the final model. The primary concerns with the model were low sample size and a lack of predictions in inshore regions due poor sampling. The ensemble model AUC for Northern Quahog was 0.73, indicating that the model has fair predictive power.

Additional Metrics

Distribution Shifts:

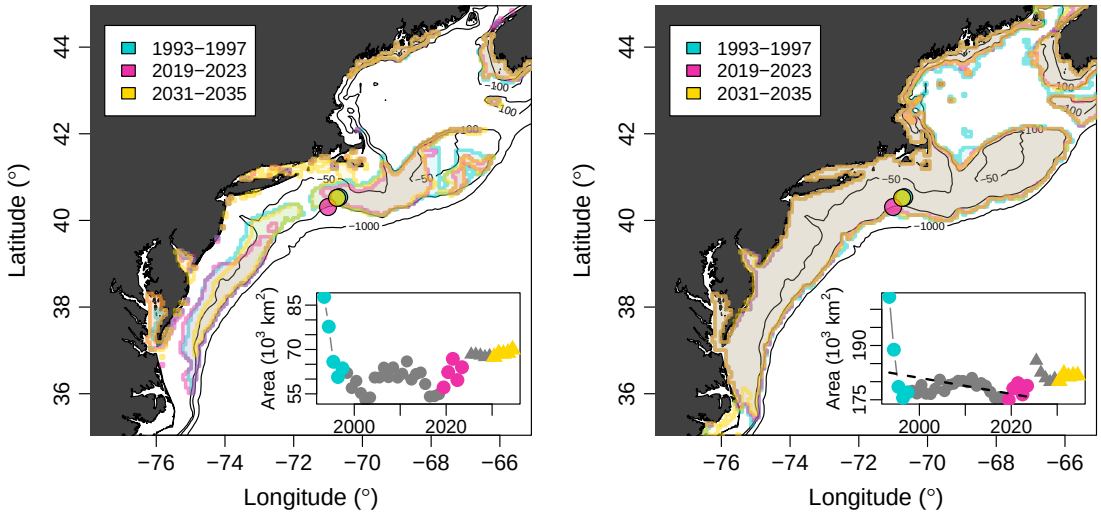


Figure 6: Distribution shifts on the Northeast Shelf for Northern Quahog. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Northern Quahog available habitat declined significantly between 1993 and 2023. Most habitat loss occurred within the Gulf of Maine and on the continental shelf south of Long Island and New England (Figure 6). The weighted centroid shifted 38 km west-southwest.

Under near future climate conditions, northern quahog habitats are not predicted to shift significantly. Distributions south of Long Island are expected to return to 1993 distributions (Figure 6). The weighted centroid is predicted to shift 32 km northeast.

In the previous Assessment, the northern quahog received a High Distributional Vulnerability Rank (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	79,139	193,388	6.86	9.30
2019-2023	37.7 km WSW	63,468 (-15,671)	177,678 (-15,710)	6.98 (+0.12)	9.46 (+0.16)
2031-2035	32.3 km NE	68,821 (+5,353)	181,775 (+4,097)	6.80 (-0.19)	9.62 (+0.16)

Directional Effect:

Experts predict that the Northern Quahog will be **negatively** impacted by climate change, with 70% certainty. Temperature and salinity are important to reproductive success and settlement – increasing temperatures outside the preferred temperature range could inhibit spawning and increased freshwater intrusions could inhibit metamorphosis (Stanley and DeWitt 1983) (Kennish and Olsson 1975). Eutrophication and ocean acidification could also impact stock fitness (Carmichael et al. 2004) (Ivanina et al. 2013). In the previous Climate Vulnerability Assessment for the Northeast shelf, experts also predicted that Northern Quahogs would be negatively impacted by climate change (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Temperature is the decisive factor for final gamete maturation in northern quahogs. Kennish and Olsson (1975) cited 21 to 25°C as the required or preferred temperature range for spawning. Metamorphosis is inhibited at salinities below 17.5 to 20 ppt (Stanley and DeWitt 1983). Increased freshwater intrusions (man-made or otherwise) have resulted in clam die offs. Eutrophication of habitat has resulted in reduced growth and health of stocks (Carmichael et al. 2004). Recent research suggests that quahogs are relatively resilient to suboxic conditions, which may help protect them from predators that are more sensitive to these conditions (Altieri and Witman 2006). Northern quahogs exposed to higher temperatures and elevated pCO^2 led to decreased shell hardness and energy deficiency (Ivanina et al. 2013). Alternatively, northern quahogs exposed to marine heat wave conditions exhibited increased absorption efficiency, respiration rate, excretion rate, and scope for growth, indicating their ability to recover from persistent or repeated marine heat waves (Liang et al. 2024).

Life History Synopsis

The northern quahog, or hard clam, is a hard shell clam inhabiting intertidal and subtidal waters from the Gulf of St. Lawrence south to Florida (Kraeuter and Castagna 2001). Spawning occurs from May through August, depending on latitude and temperature. Female hard clams require 2 - 2.5 months to spawn out completely, but the greatest release of eggs is during the initial spawning of the season (Stanley and DeWitt 1983). Eggs hatch within 12-14 hours. Larvae are planktonic until metamorphosis, which occurs within 5-7 days at 30°C, or 16-24 days at 18°C (Stanley and DeWitt 1983) (Rice and Pechenik 1992). Eggs may be carried by currents up to 25km from spawning sites. Larvae feed on phytoplankton, primarily dinoflagellates and use physical transport mechanisms to travel to the entrance of estuaries (Stanley and DeWitt 1983). The clams prefer to settle on sandy substrate with a few small rocks and/or

shells, and set more densely in sand than mud (MacKenzie 1979). Juveniles likely consume phytoplankton or small zooplankton. The adult hard clam feeds by filtering out plankton and microorganisms that are carried along the bottom by currents (Stanley and DeWitt 1983), burrowing into the sediment and leaving their siphons exposed to feed. However, adults cease filter feeding at 6°C (Kraeuter and Castagna 2001). They are preyed upon by gastropods, crabs, and starfish (Stanley and DeWitt 1983). These clams have slow growth rates, reaching maturity at 1-3 years with a maximum lifespan of 12-20 years (Bricelj 1992), although they have been found up to 46 years (Ridgway et al. 2011). There are numerous reports of isolated incidents of hard clam disease causing high mortality rates. Disease may be related to a parasite (QPX - Quahog parasite unknown). The prevalence of QPX appears to be increasing (Smolowitz et al. 1998) (Dove et al. 2004). Quahogs occur in areas of harmful algal blooms (HABs) and may act to reduce prevalence and length of HABs due to filtration. Negative effects of HABs on quahog have been noted, specifically the presence of HABs have been shown to impact growth rates (Cerrato et al. 2004). Because these clams are primarily estuarine and inhabit specific bays and coastal zones, the fishery is managed on a localized, state-by-state basis rather than by federal fishery councils. Primary management tools include minimum legal size limits (typically 50 mm), gear restrictions, and daily bag limits (North Carolina Division of Marine Fisheries (NCDMF) 2025) (New Jersey Department of Environmental Protection (NJDEP) 2014).

Supplemental Scores

Exposure Timeseries

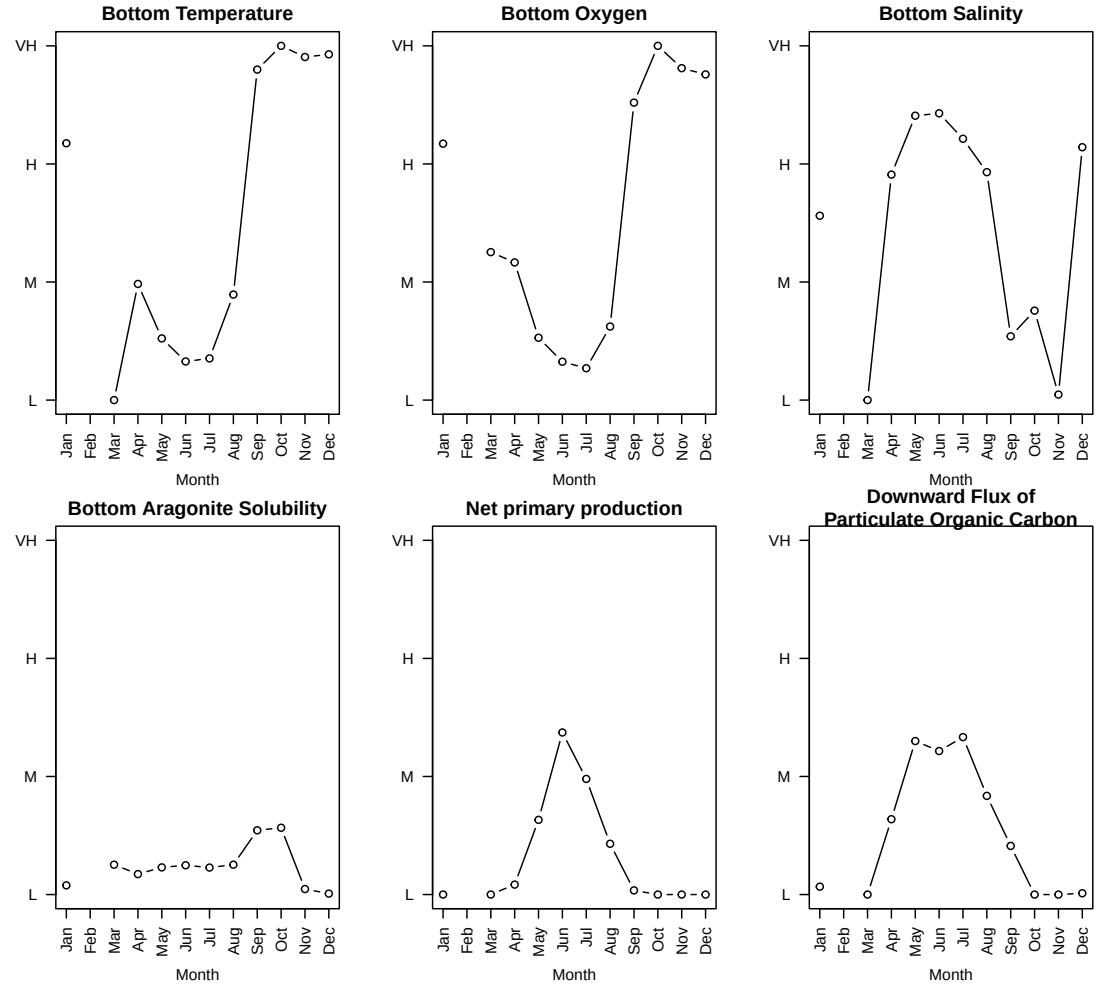


Figure 7: Monthly Timeseries of each Exposure Factor for Northern Quahog. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposures ranged from Low to High. Exposure was High from September through January (Figure 8). These patterns were driven by very high bottom temperature and oxygen exposure (Figure 7).

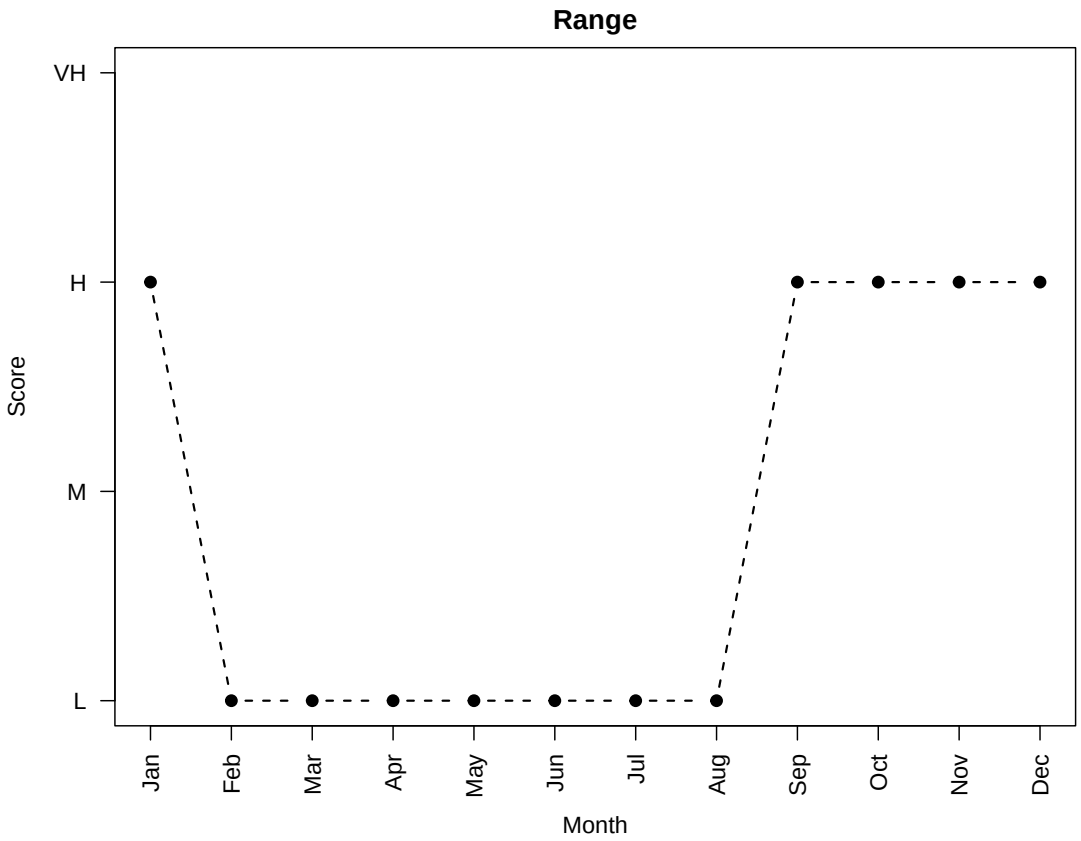


Figure 8: Timeseries of Total Exposure for Northern Quahog

Exposure due to Significant Factors

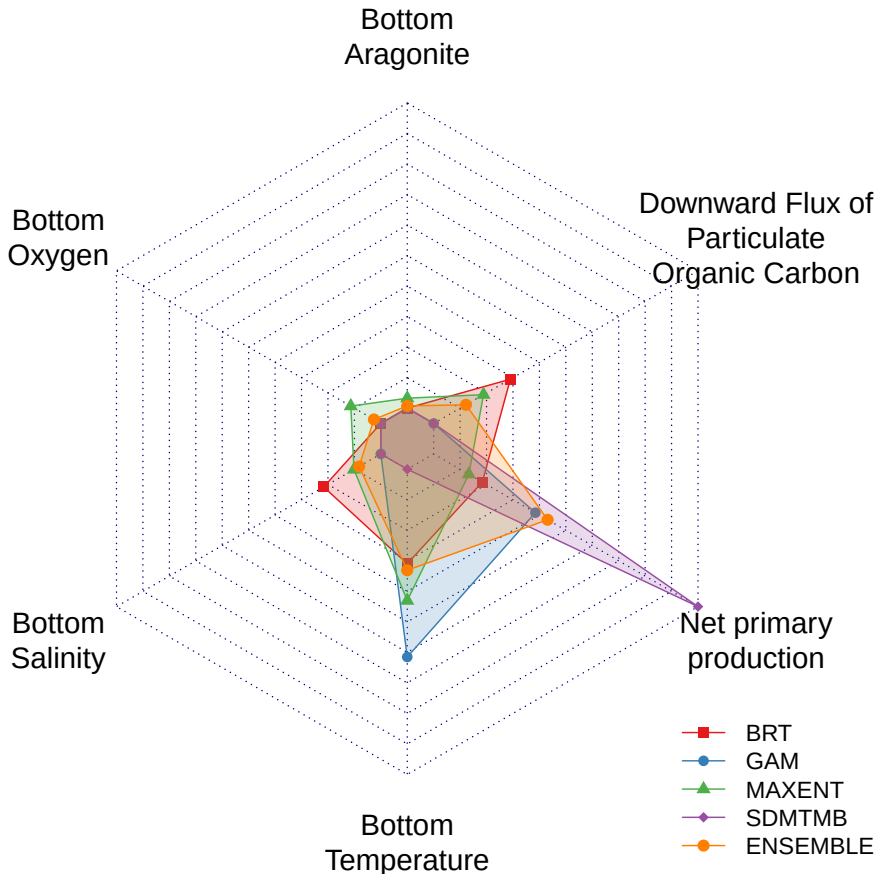


Figure 9: Dynamic Factors driving Northern Quahog distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, net primary productivity, and downward POC flux were the most important dynamic drivers of quahog distributions (Figure 9). Northern quahog are benthic filter feeders (Stanley and DeWitt 1983). Temperature is a known driver of feeding behaviors (Kraeuter and Castagna 2001). When only these factors were used to calculate exposure, mean Exposure was Low throughout its range (Figure 10, Figure 11).

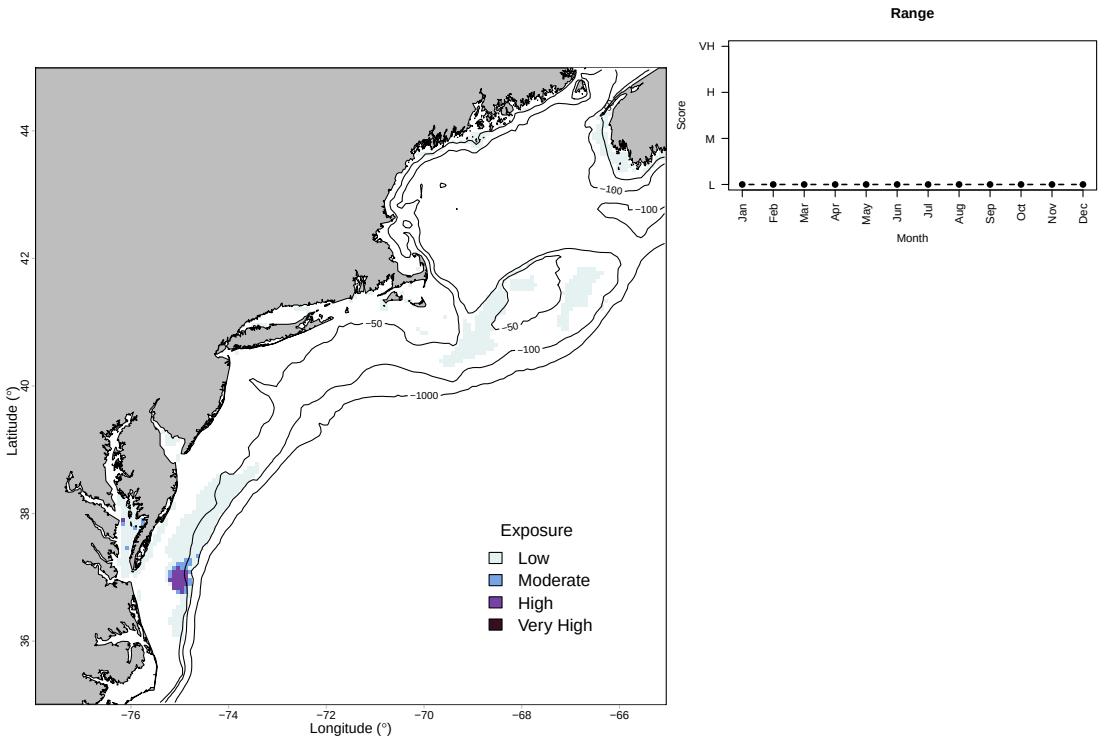
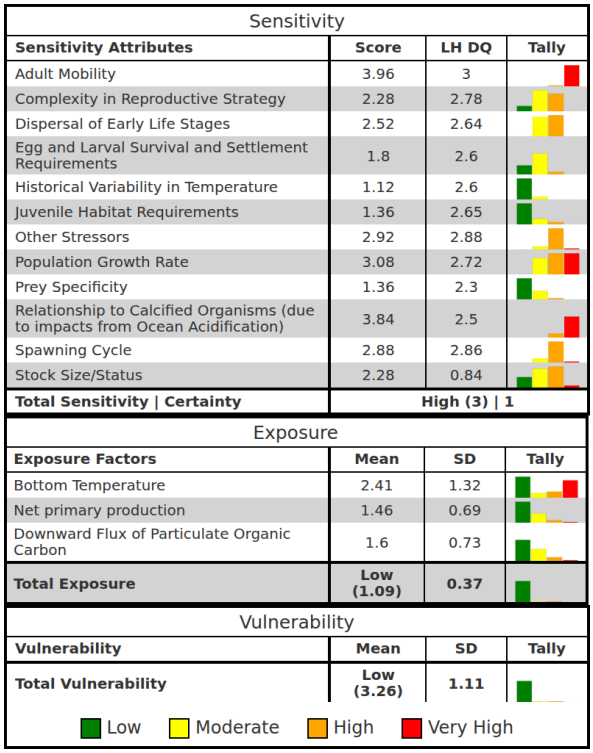


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Northern Quahog distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	3.96	3	
Complexity in Reproductive Strategy	2.28	2.78	
Dispersal of Early Life Stages	2.52	2.64	
Egg and Larval Survival and Settlement Requirements	1.8	2.6	
Historical Variability in Temperature	1.12	2.6	
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Spawning Cycle	2.88	2.86	
Stock Size/Status	2.28	0.84	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.41	1.32	
Net primary production	1.46	0.69	
Downward Flux of Particulate Organic Carbon	1.6	0.73	
Total Exposure	**Low (1.09)**	**0.37**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (3.26)**	**1.11**	

Figure 11: As in Figure 1 for Northern Quahog, but with only significant Exposure factors and the resulting Vulnerability Scores.

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